edual editorial UNIVERSIDAD DE ALMERÍA

Electronic Journal of Research in Educational Psychology ejrep

UNIVERSIDAD DE ALMERÍA

Ilustre Colegio Oficial de Psicología DE ANDALUCÍA ORIENTAL

Changes in procrastinating behaviour through intervention in university students: A systematic review

Arminda Suárez-Perdomo[1], **Zuleica Ruiz-Alfonso** [1]

[1] Department of Teaching and Educational Research, University of La Laguna, San Cristóbal de La Laguna

Spain

Correspondence: Arminda Suárez-Perdomo. Department of Teaching and Educational Research, Universidad de La Laguna, C/ Pedro Zerolo, s/n. Edificio Central. Planta 2. Apartado 456. Código postal 38200. San Cristóbal de La Laguna (Tenerife), Spain. asuper@ull.edu.es, https://orcid.org/0000-0002-6755-5284

© Universidad de Almería and Ilustre Colegio Oficial de la Psicología de Andalucía Oriental (Spain)

Abstract

Academic procrastination is a behaviour that delays the completion of tasks, time management or other academic activities. However, it is possible to develop interventions that eliminate or minimise this procrastinating behaviour. This study presents a systematic review of the literature following the PRISMA method guidelines, with the aim of identifying interventions for the reduction or elimination of procrastinating behaviours in university students. A search was conducted for records published in peer-reviewed scientific journals between 2010 and 2022. Three electronic databases were consulted. A total of ten studies that met the inclusion criteria were analysed. This review shows the main features of the included studies, with special emphasis on the changes perceived depending on the intervention modes applied: face-to-face, online and hybrid. The results revealed that face-to-face interventions showed high levels of changes perceived, as participants presented improved time management, task planning, awareness and reduced procrastination behaviour. However, although the hybrid and online interventions showed some changes in terms of the competencies identified, they did not report changes perceived in terms of changes in procrastination behaviour.

Keywords: Academic procrastination; interventions; systematic review; undergraduate.

Resumen

La procrastinación académica es un comportamiento que retrasa la realización de tareas, la gestión del tiempo u otras actividades académicas. Sin embargo, es posible desarrollar intervenciones que eliminen o minimicen esta conducta procrastinadora. Este estudio presenta una revisión sistemática de la literatura siguiendo las directrices del método PRISMA, con el objetivo de identificar intervenciones para la reducción o eliminación de las conductas procrastinadoras en estudiantes universitarios. Se realizó una búsqueda de registros publicados en revistas científicas revisadas por pares entre 2010 y 2022. Se consultaron tres bases de datos electrónicas. Se analizaron un total de diez estudios que cumplían los criterios de inclusión. Esta revisión muestra las principales características de los estudios incluidos, con especial énfasis en los cambios percibidos en función de las modalidades de intervención aplicadas: presencial, online e híbrida. Los resultados revelaron que las intervenciones presenciales mostraron altos niveles de cambios percibidos, ya que los participantes presentaron mejoras en la gestión del tiempo, la planificación de tareas, la concienciación y la reducción de la conducta de procrastinación. Sin embargo, aunque las intervenciones híbridas y en línea mostraron algunos cambios en cuanto a las competencias identificadas, no reportaron cambios percibidos en cuanto a cambios en el comportamiento de procrastinación.

Palabras Clave: Procrastinación académica, intervenciones, revisión sistemática, estudiantes universitarios.

Introduction

Procrastination is the irrational tendency or habit of unnecessarily delaying tasks that need to be completed (Ferrari et al., 2007). This tendency is a problem that can affect students' academic performance and lead to personal and emotional issues (Kim & Seo, 2015; Steel, 2007; Zacks & Hen, 2018). Scientific literature includes extensive in-depth research into the concept, causes and consequences of procrastination behaviour. (Klassen et al., 2008; Sirois & Pychyl, 2013; Steel, 2007; Steel & Ferrari, 2013). According to Steel (2007), 75% of university students procrastinate and 50% admit that they frequently procrastinate in their studies. Authors such as Zarrin et al. (2020) relate academic procrastination to fear of failure, time management difficulties, or task aversion.

The majority of university students who admit to procrastinating behaviours are characterised by a clear preference for completing tasks that provide immediate reinforcement, delaying long-term planned activities and showing a lack of self-regulation in learning (De la Fuente et al., 2021). These students tend to justify this conduct by the poor time planning and lack of interest generated by academic tasks and activities (Klassen et al., 2008). Procrastination behaviour in university students may be exacerbated by different factors such as perceived competence (Haghbin et al., 2012), fear of failure (Zhang et al., 2018), lack of self-control (Uzun et al., 2020), depression (Kinik & Odaci, 2020), low self-esteem (Duru & Balkis, 2014) or anxiety (Carden et al., 2004). In addition, procrastination hinders goal achievement and is associated with feelings of guilt, shame, and stress processes (Myrick, 2015), as well as poorer academic performance (Kim & Seo, 2015).

Considering the effects of procrastination and the importance of assessing whether these are prolonged over time (Haghbin, 2015), the need to look for ways to reduce or even overcome this behaviour becomes evident (van Eerde & Klingsieck, 2018). However, there are some controversies around the stability of procrastination behaviours that call into question the effectiveness of interventions to reduce or eliminate procrastination. For example, Gustavson et al. (2017) argue that procrastination is strongly related to awareness and impulsivity, which are considered to be stable. On the other hand, authors such as Adriaanse et al. (2010) state that procrastination is not a stable behaviour and can be modified through the formulation of a desire for change of a given behaviour. Acknowledging this premise, the person who procrastinates has the ability to change this behaviour and stop procrastinating.

There is research reporting on intervention experiences to reduce procrastination through strategies such as time management and deadline setting (Ariely & Wertenbroch, 2002), the creation of specific goal identification plans (Häfner et al., 2014), the practice of learning strategies (Tuckman & Schouwenburg, 2004), self-regulation enhancement (Steel, 2007) or cognitive-behavioural therapy (Steel, 2007; van Eerde & Klingsieck, 2018). Other interventions report effective experiences through awareness of procrastination, allowing its reduction through other aspects such as motivation, emotional control, or perceived self-efficacy (Sitzmann & Ely, 2011; van Eerde & Klingsieck, 2018).

Although interventions have traditionally been face-to-face, their practicality and cost-effectiveness are limited by multiple factors such as availability, the skill of the professionals involved, access barriers to participation, and the time investment required for the intervention process (Rozental et al., 2015). So, in recent years there has been an increase in the number of online or hybrid interventions that could help reduce the barriers of face-to-face interventions (Cerezo et al., 2020; Goda et al., 2015). Web-based courses have tools that can benefit student learning, academic performance or even reduce procrastination, as they integrate synchronous and asynchronous communication systems and offer flexibility in the distribution of responsibilities between instructor and student (Tuckman, 2007; Tuckman & Schouwenburg, 2004). The effectiveness of these courses depends to a large extent on students' adaptive use of the tools (Cerezo et al., 2015). Only a minority of students, when faced with online courses, regulate their behaviour according to the demands of the course, an adaptive pattern that enables them to achieve academic success. As noted by Azevedo & Feyzi-Behnagh (2011), procrastinating students are not appropriately focused on the task and this causes a dysregulated behaviour and need to learn techniques and tools to help them regulate this conduct and reduce their procrastination behaviour. This has led to the creation of virtual learning environments based on e-learning and focused on self-regulated learning processes. In these environments, support is provided to students in a flexible manner and with tracking and recording capabilities to gather information about the learning process. This is done, for example, through chat logs, logging of the uses of learning resources, or interaction logs (Bogarín et al., 2018). According to Jo et al. (2014) interventions in virtual learning environments could lead to increased performance and reduced procrastination, especially if there is regularity in teacher-student interaction. Bearing this in mind, online or hybrid environments could be considered as an opportunity for intervention with procrastinating students.

In contrast, other works in the literature indicate that web-based courses may increase procrastination behaviours, as these learning environments are associated with higher dropout rates resulting from study avoidance conduct and lower levels of participation (Tuckman, 2005). For example, one of the problems observed is the lack of participation in the forum during the learning process, which is positively related to procrastination and poor academic performance (Nicholas et al., 2015). Likewise, Winters et al. (2008) reported that in these environments the student must exhibit self-regulation skills to increase the likelihood of success. Authors such as Küchler et al. (2019) stated that student attrition could be one of the major problems of procrastination intervention programmes, although this lack of student adherence could be reduced with active follow-up by teachers.

As noted, the situation of procrastination is observed in the university context. Overall, students with high levels of procrastination feel more dissatisfaction, show poorer mental health, stress, social problems, and are more susceptible to illness and poorer academic performance (Höcker et al., 2013). As outlined by Küchler et al. (2019), helping college students with high procrastination levels is an important issue that requires attention. Also, added to the controversies found in the literature regarding the efficacy of interventions and their different modalities, there is a clear need for a systematic review to analyse and synthesise the characteristics and effectiveness of the results afforded by interventions in the context of academic procrastination. In this case, focusing on the university context is important, as procrastinating university students feel more dissatisfaction, show poorer mental health, stress, social problems, and are more susceptible to illness and poorer academic performance (Höcker et al., 2013). As noted by Küchler et al. (2019), helping college students with high procrastination levels is an important issue that requires attention. For these reasons, this study proposes a systematic review with the following objectives:

1. Identify the main features of interventions designed to improve procrastination behaviours in university students.

2. Examine the different intervention modalities (face-to-face, hybrid and online) and analyse which of them achieves more changes in procrastination behaviour in the university context.

Method

A systematic review was conducted to identify possible interventions designed to reduce procrastination in university students, following the guidelines set by PRISMA 2020 for conducting systematic searches and meta-analyses (Page et al., 2021). To ensure the scientific rigour and quality of the study, a series of phases were proposed: (1) The main theme of the study was presented; (2) the objectives guiding the research carried out were identified; (3) the databases and search formulae to be used were selected; (4) the diagram showing the items found that will be analysed when the selection process is completed was presented; and, (5) the results obtained from the procedure lamented were presented.

Search strategies

The search was carried out in three databases: Web of Science, Scopus and Eric. These databases were selected for their relevance and interest, as they contain collections of prestigious journals in education and psychology, among other areas, and articles of interest for the study. Before starting the search, a preliminary search was conducted on the construct of academic procrastination, as a measure for definition of the search terms and the inclusion criteria used. Subsequently, an iterative search was carried out in each database, combining the terms, and including the Boolean operators (AND, OR) with the truncated terms: Procrastination ("Academic procrastination" OR "intentional delay" OR "dilatory behaviour" OR "Postpone behaviour") AND Intervention (intervention OR "programme intervention" OR "teacher-based intervention") AND Undergraduates (Undergraduate OR "university student" OR "college student"). Table 1 shows that the search was adapted to each database in order to improve the search process.

Table 1. *Truncated search terms on each search engine*

Search engine	Truncated search terms
WOS	TS= (“Academic procrastination” OR “intentional delay” OR “dilatory behaviour” OR “Postpone behaviour”) AND TS= (intervention OR “programme intervention” OR “teacher-based intervention”) AND TS= (Undergraduate OR “university student” OR “college student”)
Scopus	(TITLE-ABS-KEY (“Academic procrastination” OR “intentional delay” OR “dilatory behaviour” OR “Postpone behaviour”) AND TITLE-ABS-KEY (intervention OR “programme intervention” OR “teacher-based intervention”) AND TITLE-ABS-KEY (Undergraduate OR “university student” OR “college student”))
ERIC	SU Descriptors= (Academic procrastination” OR “intentional delay” OR “dilatory behaviour” OR “Postpone behaviour”) AND SU Descriptors= (intervention OR “programme intervention” OR “teacher-based intervention”) AND SU Descriptors= (Undergraduate OR “university student” OR “college student”)

Eligibility criteria

A series of eligibility criteria were used to determine which studies would be included in the analysis phase. These criteria were set on the basis of the PICoS strategy (Landa-Ramírez & Arredondo-Pantaleón, 2014; Moreno & Jurado, 2021), which identifies the four elements that should be differentiated as inclusion criteria in a systematic review. Thus, the following eligibility criteria were used in this study:

- Participants Only studies conducted with university students were included, regardless of the age of the participants. Studies that focused on other types of populations were ruled out.
- Topic of interest. Studies with the objective of proposing interventions for academic procrastination were included, while those without an intervention component were excluded.
- Context. We included studies that were carried out in a university context, rejecting articles produced in other types of settings, such as social, family, or economic contexts.
- Study design. Studies that designed and evaluated an intervention were included, detailing the design drawn up and the outcomes obtained. Empirical studies that did not perform an intervention were rejected.

Two other criteria were also proposed: language, insofar as only studies in English and Spanish were included, and the publication range, from 2010 to 2022. Articles that did not meet these two final criteria were rejected.

Selection procedure

The following steps were followed for selection of the articles analysed: (1) After a preliminary reading, the search terms and inclusion and exclusion criteria were determined; (2) databases were searched and exported to the Rayyan system (Ouzzani et al., 2016), which is used to prepare systematic reviews; (3) duplicate records were identified and removed from the total number of records; (4) a selection of the resulting records was made, taking as criteria the title and/or abstract, excluding articles that did not coincide with the set objectives and inclusion criteria; and, (5) the resulting records were read for analysis.

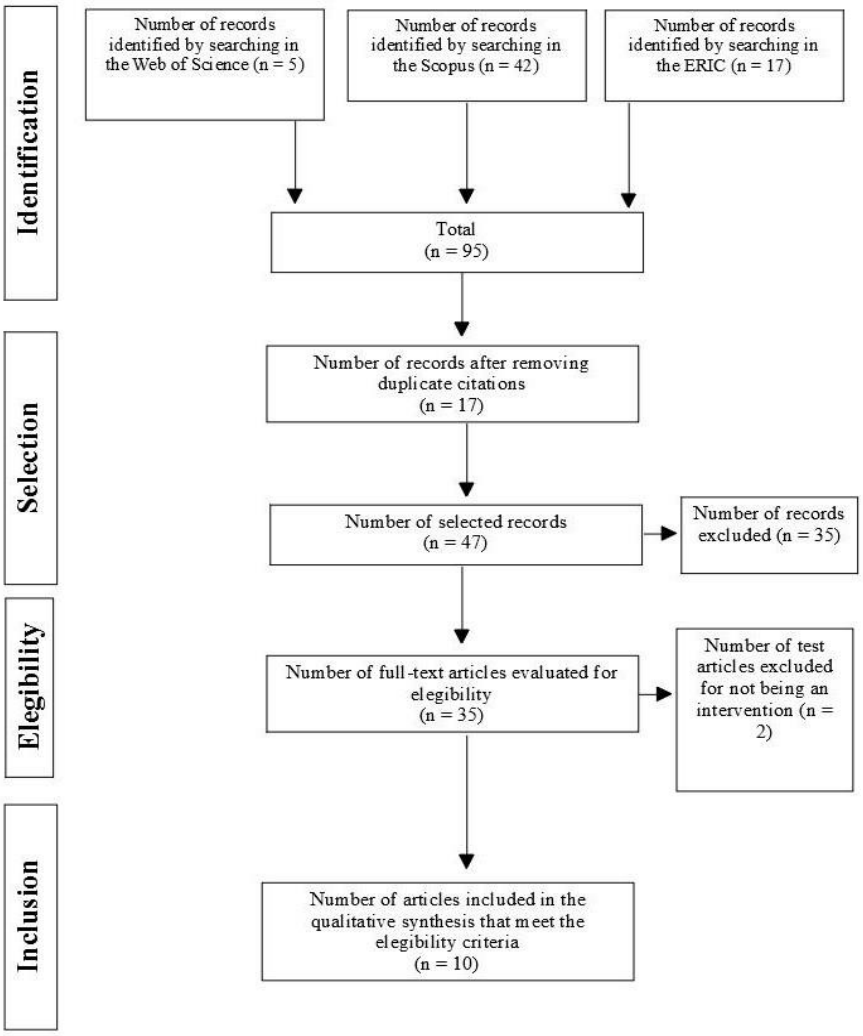


Figure 2. *Flowchart of the study selection process.*

Figure 2 shows the records found and the selection made in each of the steps, up to the items finally selected and analysed. Once the articles were selected, perused, and analysed in order to respond to the objectives of the present study.

Results

Main characteristics of the studies analysed

Ten studies were obtained as a result of the search and selection process. Table 2 shows their main features.

Table 2. *Main characteristics of the studies analysed*

Authors (year)	**Intervention modality**	**Assessment instrument**	**University (country)**	**Participants**	**Intervention duration**
1. Betul-Yilmaz (2017)	Hybrid design with 2 groups (face-to-face and online)	Academic Procrastination Scale (APS) (Çakıcı, 2003)	Yildiz Technical University (Turkey)	88	15 weeks
2. Blouin-Hudon & Pychyl (2017)	Face-to-face design with 2 groups (experimental and control)	Multidimensional Measure of Procrastination (MMoP) (Haghbin, 2015)	Carleton University (Canada)	193	Four weeks
3. Çelik & Odaci (2018)	Face-to-face therapy-based design with 2 groups (experimental and control)	Academic Procrastination Scale (APS) (Çakıcı, 2003)	Recep Tayyip Erdoğan University (Turkey)	36	10 weeks
4. Cerezo et al. (2017)	Online LMS-based design (Moodle)	Extraction of data recorded in Moodle database	University of Oviedo (Spain)	140	One semester
5. Davis & Abbit (2013)	Hybrid design (face-to-face and online)	Questionnaires via Sakai CLE 2.8 learning management system	University of Miami (USA)	124	One semester
6. Düsmez & Barut (2016)	Face-to-face design with 2 groups (experimental and control)	Academic Procrastination Scale (APS) (Çakıcı, 2003)	Giresun University (Turkey)	130	One academic year

7. Erdos-Yastibas (2020)	Face-to-face design with 2 groups (experimental and control)	Academic Procrastination Scale (APS) (Çakıcı, 2003)	Atilim University (Turkey)	64	One academic year
8. Häfner et al. (2014)	Face-to-face design with 2 groups (experimental and control)	Activity log of the selected task	University of Würzburg (Germany)	96	Four weeks
9. Hesley et al. (2021)	Face-to-face design with 2 groups (experimental and control)	Procrastination scale (ATPS, Tuckman, 1991)	Ohio State University (USA)	404	One semester
10. Toker & Avcı (2015)	Face-to-face design with 2 groups (experimental and control)	Academic Procrastination Scale (APS) (Çakıcı, 2003)	Muğla Sıtkı Koçman University (Turkey)	26	One academic year

Intervention Modality.

The majority of the studies (70%) designed interventions that were implemented in a fully face-to-face manner (Blouin-Hudon and Pychyl, 2017; Çelik & Odaci, 2018; Düsmez & Barut, 2016; Erdos-Yastibas, 2020; Häfner et al., 2014; Hesley et al., 2021; Toker & Avcı, 2015). All of these interventions were carried out using an experimental design in which the students were split into two groups:

- Control group: No intervention of any kind was carried out (Çelik & Odaci, 2018), no activity related to the modification of procrastination took place (Düsmez & Barut, 2015; Erdos-Yatisbas, 2020; Toker & Avcı, 2015), nor was any similar content developed (Blouin-Hudon & Pychyl, 2016; Häfner et al., 2014; Hesley et al., 2021).
- Experimental group: Training processes based on time management, reasons and causes of procrastination, possible solutions and irrational beliefs related to procrastination were proposed (Düsmez & Barut, 2015; Erdos-Yatisbas, 2020; Häfner et al., 2014). In other studies, they worked on awareness of the causes and consequences of procrastination (Çelik & Odaci, 2018), metacognition strategies focused on self-awareness, the development of time management strategies (Hensley et al., 2021), manipulation of self-continuity by means of mental imagery (Blouin-Hudon & Pychyl, 2017) and cognitive-behavioural training for procrastination behaviour management (Toker & Avcı, 2015).

For example, in the work by Häfner et al. (2014), an intervention was planned with two groups designated: (1) experimental training group, in which time management behaviours were practiced; (2) active control group, in which an alternative training was given with the same duration and structure comparable to those of the experimental group, but with different content (they collected and discussed time management problems). Although the authors conducted a similar intervention in both groups, the intervention with the experimental group was aimed at improving time management skills and avoiding procrastination, while in the control group only the effects of time management were explored. To this end, participants in the experimental group had to choose a task (a paper, studying for an exam, or others) that they wanted to perform in the following four weeks, recording the time spent in a diary with 20-minute time intervals. Häfner et al. (2014) provided the students with a series of steps (identifying useful next steps, anticipating possible hurdles, ways of overcoming impediments and formulating how to start working towards the objective) so that they could achieve the goals they had set. They proposed daily planning activities, setting out priorities, awareness of the time dedicated to completing the task and the implementation of specific steps to achieve the objective. In turn, the active control group collated and discussed personal time management problems. Participants talked about their problems but did not receive special guidance from the trainer or practise time management behaviours.

On the other hand, two studies designed interventions that were implemented in a hybrid manner. These interventions were either proposed with one face-to-face and one online group (Betul-Yilmuz, 2017) or used a hybrid model for all participants (Davis & Abbit, 2013). Finally, one intervention (Cerezo et al., 2017) was implemented entirely online. Specifically, Betul-Yilmaz (2017) conducted an intervention by dividing students into two groups. The first group received in-person training in a computer lab and the second received training through a distance learning scenario. Both groups received the same study plan and the same contents week by week, which also included practical activities related to procrastination. Similarly, Davis & Abbit (2013) developed a pilot course in hybrid mode, using a system of outbound text messages received by students while the intervention took place. The authors proposed a system to tackle student task avoidance by means of negative reinforcement. Thus, students were sent reminder messages about the activities until they handed them in. The plan was based on an algorithm that increased the frequency of text messages sent per day as the delivery date drew near. These deliveries followed a predetermined pattern with 13 messages per day, although the frequency was reduced to 50% the day before the delivery of the activities.

According to the authors, students were provided with a form to cancel messages if they had already completed the activities and wished to skip the messages for that week or when they wished to cancel their participation in the intervention programme.

Finally, Cerezo et al. (2017) proposed an online intervention. They created a study plan in Moodle, a learning management system (LMS), designated eTRAL Program (Cerezo et al., 2010; Núñez et al., 2011). Through this self-regulated learning programme, study techniques were offered as part of an undergraduate course. The programme lasted 11 weeks and was divided into blocks that could be accessed on the Monday of each week. Students had two weeks to complete each block, and, in them, they had to complete three mandatory tasks: a) review a theoretical content on learning strategies; b) complete a practical task related to the theoretical content; and c) contribute with a post in a social forum. The work done by the students throughout the process was automatically recorded in Moodle (date, type of action and name of the resource). With this information, the authors extracted a series of variables related to procrastination behaviour: the days that elapsed from when the block was available until the student first checked its contents, the days from when the task was visible until the student first checked the practical contents, and the days from when a block was unlocked until the student completed the task. These variables were related to student performance in the course.

Evaluation instrument.

With respect to the instruments used to analyse procrastination behaviour in five interventions (Betul-Yilmaz, 2017; Çelik & Odaci, 2018; Düsmez & Barut, 2016; Erdos-Yatisba, 2020; Toker & Avcı, 2015) the Academic Procrastination School (APS) was used Çakici, 2003). This scale is composed of 19 items (12 negative statements and 7 positive statements). “I leave my tasks/projects to the last day for trivial reasons”) with a 5-point Likert scale response system (1 = does not describe me, 5 = fully describes me). The scale consists of two factors, procrastination, and regular study habit, which measure students' tendency to procrastinate in learning experiences, preparing for exams and project development. Scores close to 95 indicate procrastination behaviour.

In turn, Hensley et al. (2021) used an estimate derived from the Tuckman Procrastination Scale (ATPS, 1991), a measure that assesses students' tendency to postpone academic work and miss deadlines on a Likert scale response system (1 = never, 5 = always). This instrument consists of 9 items divided into two factors, failure to meet deadlines (“I often

fail to complete my tasks in time") and procrastination ("I put off starting things I don't like doing").

Blouin-Hudon & Pychyl (2017) used the Multidimensional Measure of Procrastination (MMoP, Haghbin, 2015) to gauge aspects of academic procrastination associated with emotion and cognition. Specifically, the authors used behaviour-related items, although they did not specify how many items made up the instrument applied. This instrument is divided into two subscales, irrational delay ("Despite my intention to start and finish homework on time, I engage in other unnecessary activities"), and hedonistic delay ("I am not interested in starting academic tasks early because I prefer to do more fun things"), with a Likert scale response system between 1 (never) and 5 (always).

Cerezo et al. (2017) analysed students' procrastination behaviour after the intervention, extracting data recorded directly from Moodle. These data were taken as variables and organised into two groups: 1) variables related to work effort and time (time theory, time task, time forum and relevant actions as indirect indices of students' effort); and 2) procrastination-related variables (number of days students wait to review homework, forum, and theoretical content, as well as to turn in homework and post feedback).

Lastly et al. (2013) assessed the effect of text messaging on procrastination behaviour and performance. To this end, they operationally analysed the difference between the time a questionnaire was due through the Sakai CLE 2.8 LMS and the time the student started the quiz. The authors considered that the student presented low procrastination when there was a greater difference in time between the start of the questionnaire and the deadline. In contrast, they considered that the student showed high procrastination when this difference between the start of the questionnaire and the deadline was lower, i.e., the student started the questionnaire later, closer to the completion date.

University and Participants.

Five of the interventions were carried out at Turkish universities, two at universities in the USA, one in Spain, one in Canada and one in Germany. Regarding the procedure for participating in the interventions, in five of the interventions the students attended on a voluntary basis (Betul-Yilmez, 2017; Düsmez & Barut, 2016; Davis & Abbit, 2013; Häfner et al., 2014; Tecker & Avci, 2015) and in four there was an assignment of students who,

previously, were enrolled in courses established in the curricula (Blouin-Hudo & Pychyl, 2017; Cerezo et al., 2017; Erdos-Yastibas, 2020; Hesley et al., 2021). Finally, in one intervention (Çelik & Odaci, 2018), the la Academic Procrastination Scale (APS; Çakıcı, 2003) was given to a group of voluntary students, but only those students who had achieved higher scores for procrastination took part. Regarding degrees, two studies did not specify participants' field of study (Erdos-Yatisbas, 2020; Häfner et al., 2014).In four studies, the intervention targeted students in education (Betul-Yilmaz, 2017; Çelik & Odaci, 2017; Davis & Abbit, 2013; Toker & Avci, 2015). Three studies focused on interventions with students in psychology (Blouin-Hudon & Pychyl, 2017; Cerezo et al., 2017; Düzmen & Barut, 2016); and Hesley et al. (2021) conducted the intervention with students majoring in biology.

Intervention Duration.

Regarding the duration of the intervention, three took place during one academic year (Düsmez & Barut, 2016; Erdos-Yatisbas, 2020; Toker & Avci, 2015), three had a duration of one semester (Davis & Abbit, 2013; Cerezo et al., 2017; Hesley et al., 2021) and the rest were conducted in different weeks. For example, Betul-Yilmaz's (2017) intervention had a duration of 15 weeks, Çelik and Odaci's (2018) 10 weeks, and both Häfner et al. (2014) and Blouin-Hudo and Pychyl (2017) carried out interventions lasting four weeks.

Changes according to the modality of intervention on procrastination behaviour

Table 3 presents the results of interventions to improve procrastination behaviour. In the majority of cases, the results reported by the authors show the changes efficacy following the intervention process. However, differences can be observed in terms of the intervention modality. The findings from each of the interventions are shown below, according to modality, in terms of perceived changes in procrastination behaviour.

Table 3. *Results of Interventions to improve Procrastination Behaviour*

Authors (year)	**Modality**	**Time management**	**Task planning**	**Procrastination behaviour awareness**	**Procrastinator behaviour reduction**	**Effectiveness**
Betul-Yilmaz (2017)	Hybrid	YES	YES	-	YES	-
Blouin-Hudon & Pychyl (2017)	Face-to-face	YES	YES	YES	-	YES
Çelik & Odaci (2018)	Face-to-face	YES	-	YES	YES	YES
Cerezo et al. (2017)	Online	-	YES	YES	-	-

Davis & Abbit (2013)	Hybrid	-	-	YES	YES	-
Düsmez & Barut (2016)	Face-to-face	YES	-	YES	YES	YES
Erdos-Yastibas (2020)	Face-to-face	YES	-	YES	YES	YES
Häfner et al. (2014)	Face-to-face	YES	YES	-	-	YES
Hesley et al. (2021)	Face-to-face	YES	YES	-	YES	YES
Toker & Avcı (2015)	Face-to-face	-	YES	YES	YES	YES

Face-to-Face Intervention Modality.

Seven interventions based on this modality were identified. Following Blouin-Hudon and Pychyl (2017), procrastination behaviours are characterised, in part, by the lack of connection between the "future self" and the "current self". Following this postulate, in their intervention they observed that the mental images of the participants in the experimental group had a stronger connection with their "future selves". However, they could not empirically decipher why experimental group participants felt more connected to their "future self" than participants in the control group. The authors provided a plausible explanation for these changes, identifying those participants in the experimental group experienced an increase in cognitive flexibility, which entails greater ability to experience different aspects of an idea or image. According to the authors, these results demonstrate that the vividness of mental imagery plays a very important role in fostering a connection to the "future self" and facilitates greater cognitive flexibility. This leads to changes in planning and time management and results in an increased awareness of procrastination behaviours.

In turn, Çelik and Odaci (2018) observed after the intervention that the experimental group participants showed greater awareness of their procrastination behaviours, knowing, and analysing the causes of procrastination and the possibilities of reducing this behaviour. Participants also became aware of the importance of time management and its effective use. In addition, they developed motivational strategies to ensure success in academic life, positive self-perception, self-monitoring, assessment, and self-regulation. According to the authors, this led to a reduction in procrastination behaviours, in contrast to the control group, which did not show significant changes.

In the intervention carried out by Düzmez and Barut (2016), the authors stated that students in the experimental group showed increased awareness of procrastination behaviours,

and that their formative actions were rational, realistic, and pragmatic with respect to their preferences about effective time management, and that rational beliefs were ultimately effective in reducing procrastination behaviour.

Erdos-Yatisbas (2020), on the other hand, reported no statistically significant differences in perceived levels of time control awareness and effective task and exam resolution in the experimental group, although they did observe a reduction with respect to levels of academic procrastination. Moreover, most of the participants perceived that interventions based drawing up a checklist could help them develop time management awareness, schedule their time, and perform their tasks in an organised way.

Similarly, Häfner et al. (2014) evidenced that the experimental group showed greater control and time management, developing a more equitable distribution of work overtime and demonstrating greater efficiency in completion of the activities. In addition, they developed strategies to achieve their objectives, identifying the most useful steps and anticipating possible obstacles and ways to overcome impediments. The authors also identified that students in the experimental group were able to draw up and implement a daily plan with a collection of tasks, setting priorities, the time they needed to complete the tasks, and the implementation of concrete steps to achieve their goals. However, although the experimental group participants showed a higher level of awareness of time management, it did not materialise in their behaviour, so they did not show changes in their time management.

In the intervention carried out by Hesley et al. (2021), they reported that students in the experimental group demonstrated a greater commitment to obtaining a university degree. Although no significant differences were observed in terms of motivation and strategic commitment, participation in the experimental group was associated with enhanced performance and greater benefit in improving time management, task planning and reducing procrastination behaviours, considering that the intervention had high efficacy.

Finally, in the intervention conducted by Toker and Avci (2015), a decrease in the procrastination scores of experimental group students was observed, while no significant changes were observed in the control group. The authors also reported cognitive, affective, and behavioural changes that allowed the development of practical solutions that might have led to procrastination behaviours in the past. In addition, the authors related the task planning taught

in the programme to awareness of false automatic thoughts and dysfunctional beliefs related to tasks or assessment.

To summarize, seven studies were based on the face-to-face intervention modality (Table 3). Across all interventions, it was observed that undergraduates improved time management skills and increased awareness of procrastination behaviour. Additionally, the interventions showed a reduction in procrastination behaviours among participants, indicating their effectiveness.

Hybrid Intervention Modality.

Two interventions based on the hybrid modality (combination of online and face-to-face) were identified. On the one hand, Betul-Yilmaz (2017) observed that the environment determined student learning, identifying that the group that received the online intervention showed higher levels of procrastination and low quality of work by not attributing importance to task deadlines and time management. The procrastination behaviour of this group was reduced as exam dates approached. In both cases, the author observed a negative correlation between the level of procrastination and the quality of work in both environments, but especially in the online environment. In contrast, academic procrastination and test scores in the intervention correlated with each other in the face-to-face setting. It was observed that the students in the online group had poorer time management in the learning process and took the training process less seriously. The presence of exams was also found to contribute to the reduction of procrastination behaviours. However, no clear efficacy was observed in the intervention process implemented.

In the case of the intervention by Davis and Abbit (2013), the authors observed a positive effect on the SMS system elaborated in the intervention in terms of procrastination levels, although they could not determine the effect on academic performance. Participants indicated the usefulness of including reminders in SMS to reduce procrastination, improve academic performance and self-regulation, considering it a good stimulus to manage time more effectively. However, the authors pointed out the impossibility of considering the effectiveness of the SMS system in conjunction with the intervention, as they failed to link it to performance improvement in a statistical manner. However, the system's potential to help students become aware of procrastination behaviours and their possible reduction could be observed.

To sum up, two interventions were based on the hybrid intervention modality (Table 3). In this case, both interventions showed that undergraduates displayed a significant decrease in procrastination behaviour. Moreover, Betul-Yilmaz (2017) observed that the participants showed greater time management and task planning.

Online Intervention Modality.

One intervention (Cerezo et al., 2017) based on the online modality was identified. The authors noted that the intervention highlighted the influence of time management in virtual learning environments, especially on academic performance. The study identifies how procrastination behaviours can lead to poor academic outcomes while learning in an LMS. It was found that if students were unable to manage time, they tended to have lower performance, as they had less time to complete the proposed work and in consequence, they obtained a lower score. Thus, they determined in the intervention the need for early detection of procrastination behaviours and lack of self-regulation in order to provide specific support and adapt the interventions to this type of students. However, the intervention did not show significant results in terms of time management skills.

To sum up, one intervention was based on the online intervention modality (Table 3). In particular, in the intervention made by Cerezo et al. (2017), the authors observed that the participants improved the task planning and increased the procrastination behaviour awareness.

Discussion

The objectives of this systematic review were to identify the main features of interventions designed to improve procrastination behaviours in university students and to examine the changes perceived according to different intervention modalities. Ten articles were identified that met the eligibility criteria. Three intervention modalities were observed: face-to-face, online and hybrid. Thereafter, the results reported by the authors were analysed to determine students' capacity to change after the interventions took place and to consider them effective. The following is a description of the evidence found according to the characteristics of the interventions and different modalities.

Evidence found regarding the characteristics of the analysed articles

All the interventions included in this review shared the aim of reducing procrastination behaviour, although there was diversity in the way this was done. Three types of intervention modes were identified - face-to-face, hybrid and online - which differed, in turn, in the designs developed. In some cases, face-to-face experimental studies were designed (e.g.: Düsmez & Barut, 2016; Häfner et al., 2014) and another is, interventions in virtual learning environments, with the particularity of offering hybrid teaching (Davis & Abbit, 2013) or a fully online intervention (Cerezo et al., 2017).

Variability was also observed in the way authors gathered information on student procrastination. We identified the use of self-report questionnaires submitted before and after the intervention (Betul-Yilmaz, 2017; Çelik & Odaci, 2018; Düsmez & Barut, 2016) and data extraction from online platforms (Cerezo et al., 2017; Davis & Abbit, 2013).

Evidence found on the modality of each intervention

Most of the articles found reported a face-to-face intervention, showing positive results in behaviour change and procrastination awareness (Blouin-Hudon & Pychyl; 2017; Düsmez & Barut, 2016; Häfner et al., 2014). Although the presence of alternative interventions (online or hybrid) showed the opportunities offered by ICT and e-learning in the educational context (European Commission, 2014), it was identified that the results reported by these studies were not as positive as those of traditional interventions. Only in one study (Davis & Abbit, 2013) was a reduction in procrastination behaviour determined. Interventions in web-based learning spaces involve many challenges for students, who have to decide what, when how and for how long to learn (Sánchez-Santillán et al., 2016).

Thus, a clear effectiveness of face-to-face interventions is observed, reporting positive results in terms of time management, task planning, awareness and reduction of procrastination behaviour (Blouin-Hudos & Pychyl, 2017; Häfner et al., 2014; Hesley et al., 2021). However, most hybrid (Betul-Yilmaz, 2017) and online (Cerezo et al., 2017) interventions do not report these results or identify significant changes in students, which leads to the consideration that these interventions lack effectiveness. As stated by Tuckman (2005), online and hybrid learning environments are associated with a higher dropout rate, which could be related to higher levels of procrastination.

Along these lines, Cerezo et al. (2016) found that the effectiveness of virtual learning environments depends to a large extent on the adaptive use of the tools by the learners. As stated by Azevedo and Feyzi-Behnagh (2011), students who procrastinate in these environments have difficulties in becoming task-oriented, which reflects dysregulated behaviour and fosters the idea of students' handicap in adapting their behaviour to the demands of these environments. According to Cerezo et al. (2020), self-regulated learning processes are particularly important in hybrid and online learning environments, as learners are often asked to complete learning tasks with little or no support, requiring a high degree of self-regulation. In conclusion, we see that the results obtained by students in the hybrid and online interventions, although positive, suggest that students have not successfully adapted their behaviour to the self-regulatory demands of e-learning environments (Azevedo & Feyzi-Behnagh, 2011).

Nevertheless, the existence of interventions in alternative environments to the traditional settings demonstrates the importance being acquired by virtual learning environments. These systems have tracking and logging capabilities for gathering different types of data (Bogarín et al., 2018), which opens up a world of possibilities to intervene in education in general and procrastination in particular. In addition, to expand scenarios that allow the flexibility of learning spaces and adaptation to the particularities of each student and situation.

Limitations of the studies found

The main limitation found in the studies analysed refers to the lack of information reported by some of them, making it difficult to fully understand the intervention and, above all, to understand the results obtained. For example, interventions using the Academic Procrastination Scale (APS; Çakici, 2003) did not provide examples of the items that made up said instrument. In those studies, a series of negative and positive statements were pinpointed, but the treatment of the reverse questions was not identified. This makes it difficult to ascertain the effectiveness of the interventions and limits us to relying on the information reported by the authors, as the processing and interpretation of the data obtained is not known and less evidence is offered than in other studies in which the procedure carried out to obtain the results is described in detail.

Limitations of the Review Process Carried out

Bearing in mind that the aim of the present study was to identify interventions designed to reduce procrastination behaviour in university students, the main limitation of this systematic review refers to the exclusion of interventions conducted in other contexts such as the workplace (e.g., Rozental & Carlbring, 2013), which could provide relevant results for the design of effective interventions. Further research could consider the results of these studies and design interventions applied in the university context.

Practical implications and future prospects

The results of the review open a debate on the capacity for change in the procrastinating behaviour of university students in different learning environments, both face-to-face and virtual (hybrid and online). This leads to the need for further research on the effectiveness of experiences in alternative settings, to understand what the needs and gaps are, in order to address them and improve the student experience in reducing procrastinating behaviour and enhancing academic performance.

The need to assess the learning processes developed, beyond the use of quantitative instruments, is also identified. Allowing the incorporation of qualitative instruments, such as diaries, unstructured interviews, and focus groups, to capture the results beyond the implementation of self-reported questionnaires, in order allow for a deeper and richer analysis of the reality of student learning and experience.

In summary, although there is some variability in the modalities of intervention addressing procrastination behaviour in university students, the results obtained by the included studies reveal that interventions showing greater impact and change in students, and especially effectiveness in the process, are those conducted in a more traditional context. In other words, hybrid and online interventions demonstrate certain interesting changes but not as profound, hindering students in effectively reducing procrastinative behavior.

Funding

Zuleica Ruiz-Alfonso participation in this research was supported by the Spanish Ministry of Science and Innovation under a Juan de la Cierva Postdoctoral Grant [IJC2020-045247-I]

References

Adriaanse, M. A., Oettingen, G., Gollwitzer, P. M., Hennes, E. P., De Ridder, D. T., & De Wit, J. B. (2010). When planning is not enough: Fighting unhealthy snacking habits by mental contrasting with implementation intentions (MCII). European Journal of Social Psychology, 40(7), 1277-1293. https://doi.org/10.1002/ejsp.730

Ariely, D., & Wertenbroch, K. (2002). Procrastination, deadlines, and performance: Self-control by precommitment. Psychological science, 13(3), 219-224. https://doi.org/10.1111/1467-9280.00441

Azevedo, R., & Feyzi-Behnagh, R. (2011). Dysregulated learning with advanced learning technologies. Journal of e-Learning and Knowledge Society, 7(2). https://doi.org/10.20368/1971-8829/517

Azevedo, R., Witherspoon, A. M., Chauncey, A., Burkett, C., and Fike, A. (2009). MetaTutor: a metacognitive tool for enhancing self-regulated learning, in Paper Presented at the AAAI Fall Symposium: Cognitive and Metacognitive Educational Systems, Arlington, VA. https://www.aaai.org/ocs/index.php/FSS/FSS09/paper/viewPaper/995

*Beltul-Yilmaz, M. B. (2017). The Relation between Academic Procrastination of University Students and Their Assignment and Exam Performances: The Situation in Distance and Face-to-Face Learning Environments. Journal of Education and Training Studies, 5(9), 146-157. https://doi.org/10.11114/jets.v5i9.2545

*Blouin-Hudon, E. M. C., & Pychyl, T. A. (2017). A mental imagery intervention to increase future self-continuity and reduce procrastination. Applied Psychology, 66(2), 326-352. https://doi.org/10.1111/apps.12088

Bogarín, A., Cerezo, R., & Romero, C. (2018). Discovering learning processes using inductive miner: A case study with learning management systems (LMSs). Psicothema, 30(3), 322-329, https://doi.org/10.7334/psicothema2018.116

Cakıcı, D. C. (2003). An examination of the general procrastination behavior and academic procrastination behavior in high-school and university students. Unpublished master' s thesis, Ankara: Ankara University.

Carden, R., Bryant, C., & Moss, R. (2004). Locus of control, test anxiety, academic procrastination, and achievement among college students. Psychological reports, 95(2), 581-582. https://doi.org/10.2466/pr0.95.2.581-582

*Çelik, Ç., & Odacı, H. (2018). Psycho-educational group intervention based on reality therapy to cope with academic procrastination. Journal of Rational-Emotive & Cognitive-Behavior Therapy, 36(3), 220-233. https://doi.org/10.1007/s10942-017-0283-1

Cerezo, R., Bernardo, A., Esteban, M., Sánchez, M., & Tuero, E. (2015). Programas para la promoción de la autorregulación en educación superior: un estudio de la satisfacción diferencial entre

metodología presencial y virtual. European Journal of Education and Psychology, 8(1), 30-36. https://doi.org/10.1016/j.ejeps.2015.10.004

Cerezo, R., Bogarín, A., Esteban, M., & Romero, C. (2020). Process mining for self-regulated learning assessment in e-learning. Journal of Computing in Higher Education, 32(1), 74-88. https://doi.org/10.1007/s12528-019-09225-y

*Cerezo, R., Esteban, M., Sánchez-Santillán, M., & Núñez, J. C. (2017). Procrastinating behavior in computer-based learning environments to predict performance: A case study in Moodle. Frontiers in psychology, 8, 1403. https://doi.org/10.3389/fpsyg.2017.01403

*Davis, D. R., & Abbitt, J. T. (2013). An Investigation of the Impact of an Intervention to Reduce Academic Procrastination Using Short Message Service (SMS) Technology. Journal of Interactive Online Learning, 12(3). https://www.ncolr.org/jiol/issues/pdf/12.3.1.pdf

De la Fuente, J., Sander, P., Garzón-Umerenkova, A., Vera-Martínez, M. M., Fadda, S., & Gaetha, M. L. (2021). Self-regulation and regulatory teaching as determinants of academic behavioral confidence and procrastination in undergraduate students. Frontiers in Psychology, 12. https://doi.org/10.3389/fpsyg.2021.602904

Duru, E., & Balkis, M. (2014). The roles of academic procrastination tendency on the relationships among self-doubt, self-esteem and academic achievement. Egitim ve Bilim, 39(173). https://acortar.link/kR5hAK

*Düsmez, İ., & Barut, Y. (2016). Rational Emotive Behavior Based on Academic Procrastination Prevention Training Programme of Effectiveness. Participatory Educational Research, 3(1), 1-13. https://doi.org/10.17275/per.15.36.3.1

*Erdos-Yastibas, A. E. (2020). Understanding and Dealing with Academic Procrastination. i-Manager's Journal on English Language Teaching, 10(1), 1. https://doi.org/10.26634/jelt.10.1.16559

Ferrari, J. R., Diaz-Morales, J. F., O'Callaghan, J., Diaz, K., & Argumedo, D. (2007). Frequent behavioral delay tendencies by adults: International prevalence rates of chronic procrastination. Journal of Cross-Cultural Psychology, 38(4), 458-464. https://doi.org/10.1177/0022022107302314

Goda, Y., Yamada, M., Kato, H., Matsuda, T., Saito, Y., & Miyagawa, H. (2015). Procrastination and other learning behavioral types in e-learning and their relationship with learning outcomes. Learning and Individual Differences, 37, 72-80. https://doi.org/10.1016/j.lindif.2014.11.001

Gustavson, D. E., du Pont, A., Hatoum, A. S., Rhee, S. H., Kremen, W. S., Hewitt, J. K., & Friedman, N. P. (2017). Genetic and environmental associations between procrastination and internalizing/externalizing psychopathology. Clinical Psychological Science, 5(5), 798-815. https://doi.org/10.1177/2167702617706084

*Häfner, A., Oberst, V., & Stock, A. (2014). Avoiding procrastination through time management: An experimental intervention study. Educational Studies, 40(3), 352-360. https://doi.org/10.1080/03055698.2014.899487

Haghbin, M. (2015). Conceptualization and operationalization of delay: Development and validation of the multifaceted measure of academic procrastination and the delay questionnaire (Doctoral dissertation, Carleton University).

*Hensley, L., Kulesza, A., Peri, J., Brady, A. C., Wolters, C. A., Sovic, D., & Breitenberger, C. (2021). Supporting Undergraduate Biology Students' Academic Success: Comparing Two Workshop Interventions. CBE—Life Sciences Education, 20(4), ar60. https://doi.org/10.1187/cbe.21-03-0068

Höcker, A., Engberding, M., & Rist, F. (2013). Prokrastination: ein manual zur behandlung des pathologischen aufschiebens (Vol. 69). Hogrefe Verlag.

Jo, I., Kim, D, & Yoon, M (2014). Analyzing the log patterns of adult learners in LMS using learning analytics. In Proceedings of the Fourth International Conference on Learning Analytics And Knowledge (pp. 183–187). ACM.

Klassen, R. M., Krawchuk, L. L., & Rajani, S. (2008). Academic procrastination of undergraduates: Low self-efficacy to self-regulate predicts higher levels of procrastination. Contemporary Educational Psychology, 33(4), 915-931. https://doi.org/10.1016/j.cedpsych.2007.07.001

Kim, K. R., & Seo, E. H. (2015). The relationship between procrastination and academic performance: A meta-analysis. Personality and Individual Differences, 82, 26-33. https://doi.org/10.1016/j.paid.2015.02.038

Küchler, A. M., Albus, P., Ebert, D. D., & Baumeister, H. (2019). Effectiveness of an internet-based intervention for procrastination in college students (StudiCare Procrastination): Study protocol of a randomized controlled trial. Internet interventions, 17, 100245. https://doi.org/10.1016/j.invent.2019.100245

Landa-Ramírez, E., & Arredondo-Pantaleón, A. (2014). Herramienta PICO para la formulación y búsqueda de preguntas clínicamente relevantes en la psicooncología basada en la evidencia. Psicooncología, 11. https://doi.org/10.5209/rev_PSIC.2014.v11.n2-3.47387

Myrick, J. G. (2015). Emotion regulation, procrastination, and watching cat videos online: Who watches Internet cats, why, and to what effect? Computers in human behavior, 52, 168-176. https://doi.org/10.1016/j.chb.2015.06.001

Moreno, A. G., & Jurado, M. D. M. M. (2021). Una revisión sistemática sobre las variables relacionadas con las habilidades sociales y la creatividad en el periodo de la adolescencia. Apuntes de Psicología, 39(3). https://www.apuntesdepsicologia.es/index.php/revista/article/view/907

Nicholas, J., Larsen, M. E., Proudfoot, J., & Christensen, H. (2015). Mobile apps for bipolar disorder: a systematic review of features and content quality. Journal of medical Internet research, 17(8), e4581. https://doi.org/10.2196/jmir.4581

Núñez Pérez, J. C., Cerezo Menéndez, R., Bernardo Gutiérrez, A. B., Rosário, P. J. S. L. D. F., Valle Arias, A., Fernández Alba, M. E., & Suárez Fernández, N. (2011). Implementation of training

programs in self-regulated learning strategies in Moodle format: Results of an experience in higher education. Psicothema. 23, 274–281. http://hdl.handle.net/11162/4022

Ouzzani, M., Hammady, H., Fedorowicz, Z., & Elmagarmid, A. (2016). Rayyan-a web and mobile app for systematic reviews. Systematic Reviews, 5, 210. https://doi.org/10.1186/s13643-016-0384-4

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... & Moher, D. (2021). Declaración PRISMA 2020: una guía actualizada para la publicación de revisiones sistemáticas. Revista Española de Cardiología, 74(9), 790-799. https://doi.org/10.1016/j.recesp.2021.06.016

Rozental, A., & Carlbring, P. (2013). Internet-based cognitive behavior therapy for procrastination: Study protocol for a randomized controlled trial. JMIR Research Protocols, 2(2), e2801. https://doi.org/10.2196/resprot.2801

Rozental, A., Boettcher, J., Andersson, G., Schmidt, B., & Carlbring, P. (2015). Negative effects of internet interventions: a qualitative content analysis of patients' experiences with treatments delivered online. Cognitive Behaviour Therapy, 44(3), 223-236. https://doi.org/10.1080/16506073.2015.1008033

Sanchez-Santillan, M., Paule-Ruiz, M., Cerezo, R., & Alvarez-García, V. (2016). MeL: Modelo de adaptación dinámica del proceso de aprendizaje en eLearning. Anales de Psicología/Annals of Psychology, 32(1), 106-114. https://doi.org/10.6018/analesps.32.1.195071

Sirois, F., & Pychyl, T. (2013). Procrastination and the priority of short-term mood regulation: Consequences for future self. Social and personality psychology compass, 7(2), 115-127. https://doi.org/10.1111/spc3.12011

Sitzmann, T., & Ely, K. (2011). A meta-analysis of self-regulated learning in work-related training and educational attainment: what we know and where we need to go. Psychological bulletin, 137(3), 421. https://doi.org/10.1037/a0022777

Steel, P. (2007). The nature of procrastination: a meta-analytic and theoretical review of quintessential self-regulatory failure. Psychological bulletin, 133(1), 65. https://doi.org/10.1037/0033-2909.133.1.65

Steel, P., & Ferrari, J. (2013). Sex, education and procrastination: An epidemiological study of procrastinators' characteristics from a global sample. European Journal of Personality, 27(1), 51-58. https://doi.org/10.1002/per.1851

*Toker, B., & Avcı, R. (2015). Effect of cognitive-behavioral-theory-based skill training on academic procrastination behaviors of university students. 15(5), 1157-1168. https://doi.org/10.12738/estp.2015.5.0077

Tuckman, B. W. (2007). The effect of motivational scaffolding on procrastinators' distance learning outcomes. Computers & Education, 49(2), 414-422. https://doi.org/10.1016/j.compedu.2005.10.002

Tuckman, B. W., & Schouwenburg, H. C. (2004). Behavioral Interventions for Reducing Procrastination Among University Students. https://doi.org/10.1037/10808-007

Uzun, B., LeBlanc, S., & Ferrari, J. R. (2020). Relationship between academic procrastination and self-control: the mediational role of self-esteem. College Student Journal, 54(3), 309-316. https://acortar.link/jrolF6

van Eerde, W., & Klingsieck, K. B. (2018). Overcoming procrastination? A meta-analysis of intervention studies. Educational Research Review, 25, 73-85. https://doi.org/10.1016/j.edurev.2018.09.002

Winters, F. I., Greene, J. A., & Costich, C. M. (2008). Self-regulation of learning within computer-based learning environments: A critical analysis. Educational psychology review, 20(4), 429-444. https://doi.org/10.1007/s10648-008-9080-9

Zacks, S., & Hen, M. (2018). Academic interventions for academic procrastination: A review of the literature. Journal of Prevention & Intervention in the Community, 46(2), 117-130. https://doi.org/10.1080/10852352.2016.1198154

Zhang, Y., Dong, S., Fang, W., Chai, X., Mei, J., & Fan, X. (2018). Self-efficacy for self-regulation and fear of failure as mediators between self-esteem and academic procrastination among undergraduates in health professions. Advances in Health Sciences Education, 23(4), 817-830. https://doi.org/10.1007/s10459-018-9832-3

Zarrin, S. A., Gracia, E., & Paixão, M. P. (2020). Prediction of academic procrastination by fear of failure and self-regulation. Educational Sciences: Theory & Practice, 20(3), 34-43. https://doi.org/10.12738/jestp.2020.3.003

Table for Appendix. *Sample of excluded articles from Rayyan because does not correspond to the topic of interest*

Topic of interest

1. Birol, Z. N., & Günal, Y. (2019). A Study on the Relationship between the Behaviour of the Academic Procrastination of the Psychological Counseling and Guidance Students and Their Goal Orientations and Self-Esteem= Psikolojik danismanlik ve rehberlik ögrencilerinin akademik erteleme davranisi ile basari yönelimleri ve benlik saygisi arasindaki iliskinin incelenmesi. *Pegem Journal of Education and Instruction*, *9*(2), 435-450.
2. Bolden, J., & Fillauer, J. P. (2020). "Tomorrow is the busiest day of the week": Executive functions mediate the relation between procrastination and attention problems. *Journal of American College Health*, *68*(8), 854-863.
3. Bowering, E. R., Mills, J., & Merritt, A. (2017). Learning How to Learn: A Student Success Course for at Risk Students. *Canadian Journal for the Scholarship of Teaching and Learning*, *8*(3), n3.
4. Ocak, G., & Boyraz, S. (2016). Examination of the Relation between Academic Procrastination and Time Management Skills of Undergraduate Students in Terms of Some Variables. *Journal of education and training studies*, *4*(5), 76-84.
5. Sudler, E. L. (2014). *Academic procrastination as mediated by executive functioning, perfectionism, and frustration intolerance in college students*. St. John's University (New York).
6. Strunk, K. K. (2012). *Investigating a new model of time-related academic behavior: Procrastination and timely engagement by motivational orientation*. Oklahoma State University.
7. Zhang, Y., Dong, S., Fang, W., Chai, X., Mei, J., & Fan, X. (2018). Self-efficacy for self-regulation and fear of failure as mediators between self-esteem and academic procrastination among undergraduates in health professions. *Advances in Health Sciences Education*, *23*, 817-830.
8. Balkis, M., Duru, E., & Bulus, M. (2013). Analysis of the relation between academic procrastination, academic rational/irrational beliefs, time preferences to study for exams, and academic achievement: a structural model. *European journal of psychology of education*, *28*, 825-839.
9. Klibert, J., Langhinrichsen-Rohling, J., Luna, A., & Robichaux, M. (2011). Suicide proneness in college students: Relationships with gender, procrastination, and achievement motivation. *Death Studies*, *35*(7), 625-645.
10. Perdomo, A. S., & Feliciano-Garcia, L. (2020). The influence of active procrastination: a profile on educational sciences students' academic achievement. *Bordon-Rev. De Pedagogia*, *72*, 157-170.

Received: 16-02-2023
Accepted: 18-02-2024